

M532 Spring 2026
Problem Set 3

Problem 1. The Wisconsin Breast Cancer data set is a binary classification data set used to distinguish benign from malignant breast tumors. Each sample is described by numerical features computed from digitized images of cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, and concavity. It is commonly used to test classification and feature-selection methods because the features are interpretable and the classification task is medically meaningful.

Include your formulation of the sparse SVM as a linear program in your write-up.

- a) Write a code to solve the sparse support vector machine as derived in class. You may use a package to solve the linear program.
- b) Divide your data into a training, validation and test set using a 70%, 15%, 15% split. The training set is used to compute the model. The validation set is used to determine the optimal regularization parameter C . The test set is used to evaluate the model's success.
- c) For the optimal value of C selected, plot the weights in the model ordered by their absolute magnitude. How many features can be discarded?
- d) What is your accuracy on the test set using only the selected features?
- e) Compute the confusion matrix and interpret.

A *confusion matrix* is a table that compares the true class labels with the labels predicted by a classifier. The diagonal entries count the correct classifications, while the off-diagonal entries count the classification errors. For our binary classification problem, the confusion matrix has the form

	Predicted Malignant	Predicted Benign
Actual Malignant	correct malignant	missed malignant
Actual Benign	false alarm	correct benign

Problem 2 Now classify the Wisconsin breast cancer data using Fisher discriminant analysis. Use your validation set to determine the classification threshold. Compare your results using the full feature set, and the reduced feature set found in problem 1.

Problem 3 Following the application in K-SVD: An Algorithm for Designing Overcomplete Dictionaries for Sparse Representation, Michal Aharon, Michael Elad, and Alfred Bruckstein, compute a dictionary to compress a natural image of your selection.

- a) Break your image into at least 1,000 8×8 patches.

- b) Initialized your dictionary (explore sizes 64 to 512) by selecting randomly from the (flattened) patches.
- c) Plot the error at the end of Phase II for 10 iterations using the sparsity parameter $T = 8$.
- d) Order your dictionary atoms by image variance and plot the top 12.
- e) Show your best reconstruction of the image by reassembling the patches.